

OMKARESHWARI MASHETTE

Bengaluru | omkareshwarimashette@gmail.com | 7676634872 | linkedin.com/in/omkareshwari | github.com/omkareshwari

PROFESSIONAL SUMMARY

B.E. Computer Science (Data Science) graduate with experience in Machine Learning, Data Analysis, and Generative AI applications. Skilled in Python, SQL, Scikit-learn, LangChain, and Retrieval-Augmented Generation (RAG). Completed internship and academic projects involving document retrieval, classification, NLP, data preprocessing, and model evaluation. Strong problem-solving and analytical skills with interest in Data Science, Machine Learning, and Artificial Intelligence.

EDUCATION

B.E. in Computer Science & Engineering (DS) BKIT Engineering College, BIDAR CGPA: 8.1	2022 – 2026
PUC in Science , CB GURUKUL PU College, BIDAR 68%	2020 – 2022

INTERNSHIP EXPERIENCE

GenAI Intern (RAG-Based Systems) Nayagara Technologies Limited 	Jan 2026 – Jun 2026
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- Developed a document search and question-answering system using Python, LangChain, LlamaIndex, and Azure OpenAI.
- Built data ingestion, chunking, embedding, and vector indexing pipelines using Azure AI Search, FAISS, and ChromaDB.
- Designed retrieval workflows and prompt templates to improve response accuracy and document relevance.
- Developed REST APIs and performed data preprocessing, model evaluation, and experiment tracking using FastAPI, Scikit-learn, and MLflow.

PROJECTS

GenAI Operations & Policy Assistant (RAG)

Tools: Python, Azure OpenAI, LangChain, Azure AI Search, FAISS

- Implemented a RAG-based assistant for airline operations using operational manuals and SOPs.
- Built document ingestion and retrieval pipelines for efficient information access.
- Optimized response accuracy using prompt engineering and metadata-based search.

Adverse Event Seriousness Classification

Tools: Python, Scikit-learn, XGBoost, MLflow, SQL

- Developed a supervised machine learning model to classify adverse event narratives by seriousness and priority.
- Engineered features from narrative text and structured case data to improve classification performance.
- Evaluated models using stratified validation techniques and conducted error analysis for performance improvement.
- Implemented batch scoring workflows and monitored data quality throughout the model lifecycle.

ShadowLock – AI-Enhanced Steganography Platform

Tools: Python, OpenCV, Flask, NumPy, MySQL

- Designed a secure application for hiding sensitive information within digital images.
- Applied image processing and encryption techniques for secure data transmission.
- Implemented backend services for image upload, encoding, and decoding operations.

TECHNICAL SKILLS

Programming: Python, SQL, Git

Machine Learning: Scikit-learn, XGBoost, Classification, Regression, Clustering, NLP, Model Evaluation, Statistical Analysis

Generative AI: RAG, Prompt Engineering, Embeddings, Vector Search

Frameworks: LangChain, LlamaIndex, FastAPI, Flask

Databases: MySQL, FAISS, ChromaDB, Azure AI Search

Data Analysis: Pandas, NumPy, EDA, Data Visualization, Feature Engineering, Data Preprocessing

Tools & Cloud: Azure OpenAI, Azure ML, MLflow, GitHub, Power BI

Soft Skills: Problem Solving, Communication, Team Collaboration

CERTIFICATIONS

- Programming in Python - Echo Brains | 2024
- Data Science Foundations – Infosys Springboard | 2025
- AI/ML with Data Science – Knowx innovation | Feb 2026